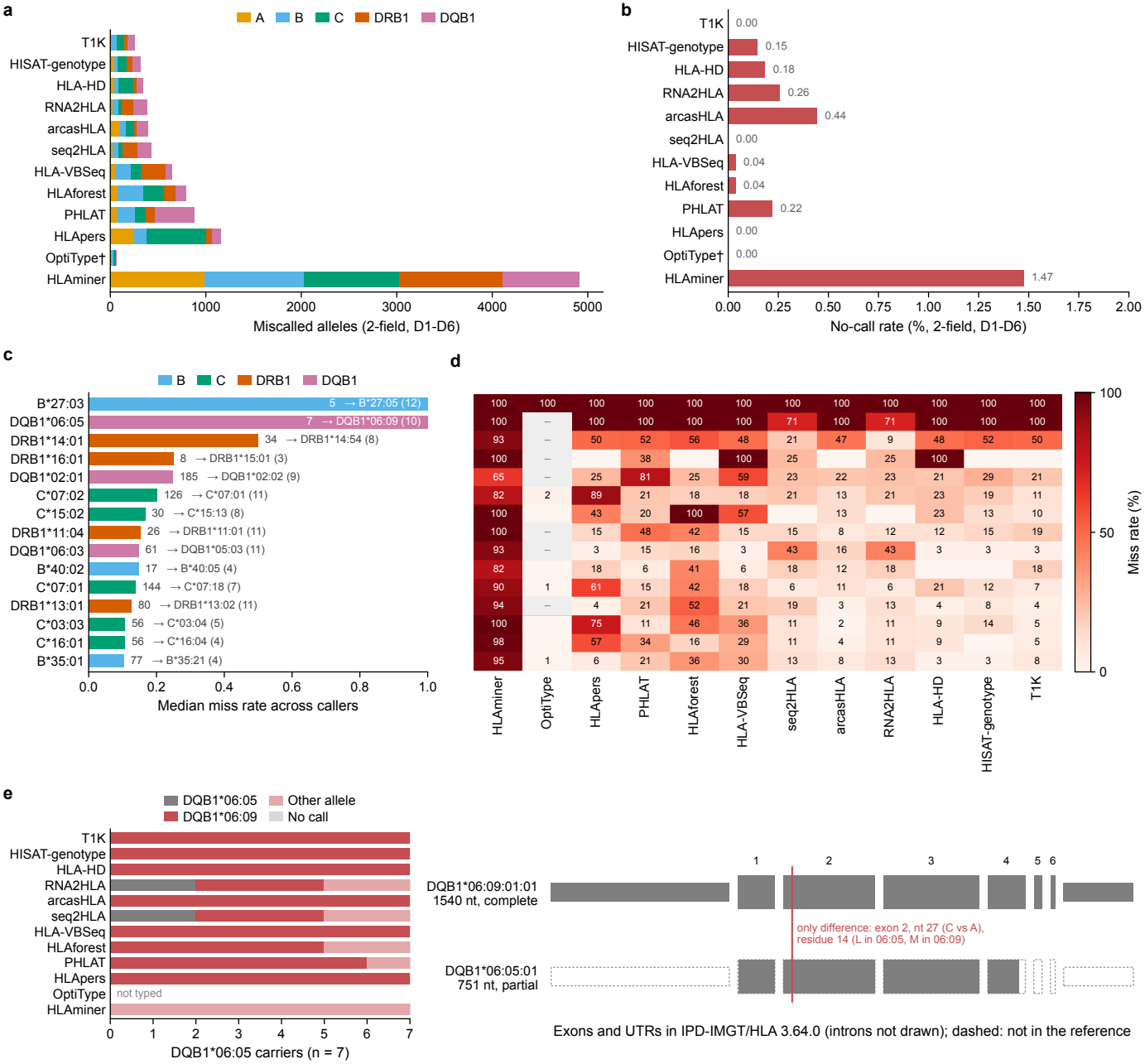
a
A
B
C
DRB1
DQB1
T1K
HISAT-genotype
HLA-HD
RNA2HLA
arcasHLA
seq2HLA
HLA-VBSeq
HLAforest
PHLAT
HLApers
OptiType†
HLAminer
0
1000
2000
3000
4000
5000
Miscalled alleles (2-field, D1-D6)
b
T1K 0.00
HISAT-genotype 0.15
HLA-HD 0.18
RNA2HLA 0.26
arcasHLA 0.44
seq2HLA 0.00
HLA-VBSeq 0.04
HLAforest 0.04
PHLAT 0.22
HLApers 0.00
OptiType† 0.00
HLAminer 1.47
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
No-call rate (%, 2-field, D1-D6)
c
B
C
DRB1
DQB1
B*27:03 5 → B*27:05 (12)
DQB1*06:05 7 → DQB1*06:09 (10)
DRB1*14:01 34 → DRB1*14:54 (8)
DRB1*16:01 8 → DRB1*15:01 (3)
DQB1*02:01 185 → DQB1*02:02 (9)
C*07:02 126 → C*07:01 (11)
C*15:02 30 → C*15:13 (8)
DRB1*11:04 26 → DRB1*11:01 (11)
DQB1*06:03 61 → DQB1*05:03 (11)
B*40:02 17 → B*40:05 (4)
C*07:01 144 → C*07:18 (7)
DRB1*13:01 80 → DRB1*13:02 (11)
C*03:03 56 → C*03:04 (5)
C*16:01 56 → C*16:04 (4)
B*35:01 77 → B*35:21 (4)
0.0
0.2
0.4
0.6
0.8
1.0
Median miss rate across callers
d
HLAminer
OptiType
HLApers
PHLAT
HLAforest
HLA-VBSeq
seq2HLA
arcasHLA
RNA2HLA
HLA-HD
HISAT-genotype
T1K
Miss rate (%)
100
50
0
e
DQB1*06:05
DQB1*06:09
Other allele
No call
T1K
HISAT-genotype
HLA-HD
RNA2HLA
arcasHLA
seq2HLA
HLA-VBSeq
HLAforest
PHLAT
HLApers
OptiType
not typed
HLAminer
0
1
2
3
4
5
6
7
DQB1*06:05 carriers (n = 7)
DQB1*06:09:01:01
1540 nt, complete
DQB1*06:05:01
751 nt, partial
1
2
3
4
5
6
only difference: exon 2, nt 27 (C vs A),
residue 14 (L in 06:05, M in 06:09)
Exons and UTRs in IPD-IMGT/HLA 3.64.0 (introns not drawn); dashed: not in the reference